

GOAL LINE Technology



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Use of technology in sports is increasing rapidly, though the sport of football has been slow to incorporate some forms of technology that other sports seem to have. While cricket and tennis implemented Hawk-Eye technique for accurate decisions, FIFA believed that the technology would hamper the pace of the game. The technology had to be incorporated ultimately due to human errors that affected the game. Goal line technology was discussed and brought into action.

Goal line technology

Goal line technology is a method used to determine when the ball has completely crossed the goal line in between goal posts and underneath the crossbar with the assistance of electronic devices and to assist the referee in awarding a goal. Objective of this technology is to support officials in their decision-making rather than replacing them. Systems such as smart-ball, Hawk-Eye and Goal Ref were taken into consideration.

International Football Association Board (IFAB) had a long trial period to ensure that they got it right before they implemented any system.

Following criteria are desired for a successful goal line system:

1. Technology should only apply to goal line decisions
2. System must be 100 per cent accurate

3. Signal sent to referee must be instantaneous

4. Signal is only communicated to match officials

Discussed below are some recent incidents that led to implementation of goal line technology.

Chelsea versus Tottenham (FA Cup Semi-Final 2012). On April 15, 2012, in Chelsea's FA Cup semi-final against Tottenham Hotspur, referee Martin Atkinson awarded Chelsea a goal resulting from a forty-ninth minute shot by Juan Mata. Atkinson ruled that the shot had crossed the line despite replays confirming that several Tottenham players had successfully blocked the effort at a point several metres in front of the goal line.

England versus Germany (World Cup Knockout Round 2010). On June 27, 2010, England was playing Germany in the knockout round of 2010 World Cup in Bloemfontein, South Africa. In the thirty-eighth minute, Frank Lampard shot the ball and it hit the underside of the crossbar, resulting in it crossing the line into the goal but bouncing back into the field of play due to the backspin. Neither the referee nor his assistants were in a position to award the goal and the play was waved on, whereas replays showed that the ball had crossed the line.

Smartball system

A promising prospect has been a smart-ball, miCoach, loaded with a sensor



Fig. 1: Wristwatch used to inform referee if the ball has crossed the line or not



Fig. 2: Tottenham players blocking the ball before it crosses the line